

LESSON 2-5 • SESSION 2 • PREREQUISITE PRACTICE B • SKILL 1 OF 2

Find the Range and IQR

Builds on Lesson 2-5 Session 1 • 6.SP.B.5c

TODAY'S LESSON

In problem 2 you find the range and the IQR of each vending machine's sales.

YOU NEED TO

Find the range and the IQR of a data set. **You will use it in Practice Form A, problems 2 and 7.**

STEPS TO FOLLOW*use these same steps on every problem*

- 1** **Order** the data. Range = max – min.
- 2** Find the quartiles, **Q1** and Q3.
- 3** **IQR** = Q3 – Q1.

WATCH ME*the steps in action***Find the range and the IQR of 22, 15, 30, 18, 27, 12, 35.****Step 1** 12, 15, 18, 22, 27, 30, 35. Range = $35 - 12 = 23$.**Step 2** Q1 = 15. Q3 = 30.**Step 3** IQR = $30 - 15 = 15$.**Answer** range **23**, IQR **15****TRY IT WITH HELP***fill in each step*

- 1.** Find the range and the IQR of 4, 7, 10, 15, 19, 21, 25.

Step 1 Range = $25 - 4 =$ _____.**Step 2** Q1 = _____. Q3 = _____.**Step 3** IQR = _____.

- 2.** Find the range and the IQR of 23, 28, 30, 36, 41, 45.

Step 1 Range = $45 - 23 =$ _____.**Step 2** Q1 = _____. Q3 = _____.**Step 3** IQR = _____.**YOUR TURN***use the steps on your own*

- 3.** Each list is in order. Write the range and the IQR.

a. 50, 52, 59, 63, 70, 72, 75 _____**b.** 8, 12, 13, 20, 26, 31, 33 _____**c.** 101, 104, 110, 118, 120 _____**d.** 66, 70, 75, 80, 92, 95 _____

Compare Box Widths

Builds on Lesson 2-4 Session 2 • 6.SP.B.4

TODAY'S LESSON

In problem 8, Bus 1's box is narrow. A narrow box means predictable.

YOU NEED TOFind two box widths and tell which group is more consistent. **You will use it in Practice Form A, problems 3 and 8.****STEPS TO FOLLOW***use these same steps on every problem*

- 1** Find each box **width**: $Q3 - Q1$.
- 2** **Compare** the two widths.
- 3** The **narrower** box is more consistent.

WATCH ME*the steps in action***Box A goes from 20 to 26. Box B goes from 18 to 34. Which is more consistent?****Step 1** A: $26 - 20 = 6$. B: $34 - 18 = 16$.**Step 2** $6 < 16$.**Step 3** Box A is narrower.**Answer** **Group A is more consistent.****TRY IT WITH HELP***fill in each step***4.** Box C goes from 30 to 44. Box D goes from 31 to 37.**Step 1** C: $44 - 30 = \underline{\hspace{2cm}}$. D: $37 - 31 = \underline{\hspace{2cm}}$.**Step 2** $\underline{\hspace{2cm}}$ is narrower.**Step 3** The more consistent group is $\underline{\hspace{2cm}}$.**5.** Box P goes from 12 to 19. Box Q goes from 40 to 46.**Step 1** P: $19 - 12 = \underline{\hspace{2cm}}$. Q: $46 - 40 = \underline{\hspace{2cm}}$.**Step 2** $\underline{\hspace{2cm}}$ is narrower.**Step 3** The more consistent group is $\underline{\hspace{2cm}}$.**YOUR TURN***use the steps on your own***6.** Find each box width ($Q3 - Q1$).**a.** Q1 12, Q3 19: $\underline{\hspace{2cm}}$ **b.** Q1 30, Q3 44: $\underline{\hspace{2cm}}$ **c.** Q1 55, Q3 61: $\underline{\hspace{2cm}}$ **d.** Q1 8, Q3 23: $\underline{\hspace{2cm}}$ **e.** Q1 71, Q3 90: $\underline{\hspace{2cm}}$ **f.** Q1 40, Q3 46: $\underline{\hspace{2cm}}$ **READY CHECK***a problem like the one in today's lesson***7.** Box A is 6 wide. Box B is 15 wide. Which is true?**A** A is more spread out.**B** B is more consistent.**C** A is more consistent.**D** They have the same spread.**CHECK YOURSELF**☐ I can do this on my own☐ I need a little help☐ I have a question about # $\underline{\hspace{1cm}}$